
Medicinal Chemistry I

Lecture Series, 2023

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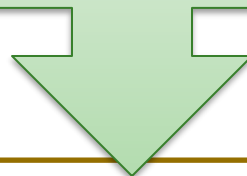
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Introduction

Definition:

Medicinal Chemistry: Is a science which deals with

Discovery and design of new and better therapeutic chemicals and development of these chemicals into new drugs.



- ❑ Their physicochemical properties & structure activity relationship which are essential for their design and therapeutic application.
- ❑ Modification of drugs into useful therapeutic agents.

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- IUPAC definition:

Medicinal Chemistry:

- It explains the design and production of compounds that can be used for the prevention, treatment or cure of human and animal diseases.
 - Concerned with intervention, discovery, design, identification and preparation of biologically active compounds.
 - ✓ Study their metabolism
 - ✓ Interpretes their mode of action at molecular level and
 - ✓ Constructs their structure-activity relationship (relationship between structure and pharmacological activity of the compounds).

- Medicinal chemistry covers the following stages:
 - In the first stage *new active substances or drugs are identified and prepared* from natural sources, organic chemical reactions or biotechnological processes. They are known as **lead molecules**.
 - The second stage is *optimization of lead structure* to improve potency, selectivity and to reduce toxicity.
 - Third stage is *development stage*, which involves optimization of synthetic route for bulk production and modification of pharmacokinetic and pharmaceutical properties of active substance to render it clinically useful.

A medicinal chemist can:

- Make new compounds (drugs).
- Determine their effect on biological processes.
- Alter the structure of the compound for optimum effect and minimum side effects.
- Study uptake, distribution, metabolism and excretion of drugs.

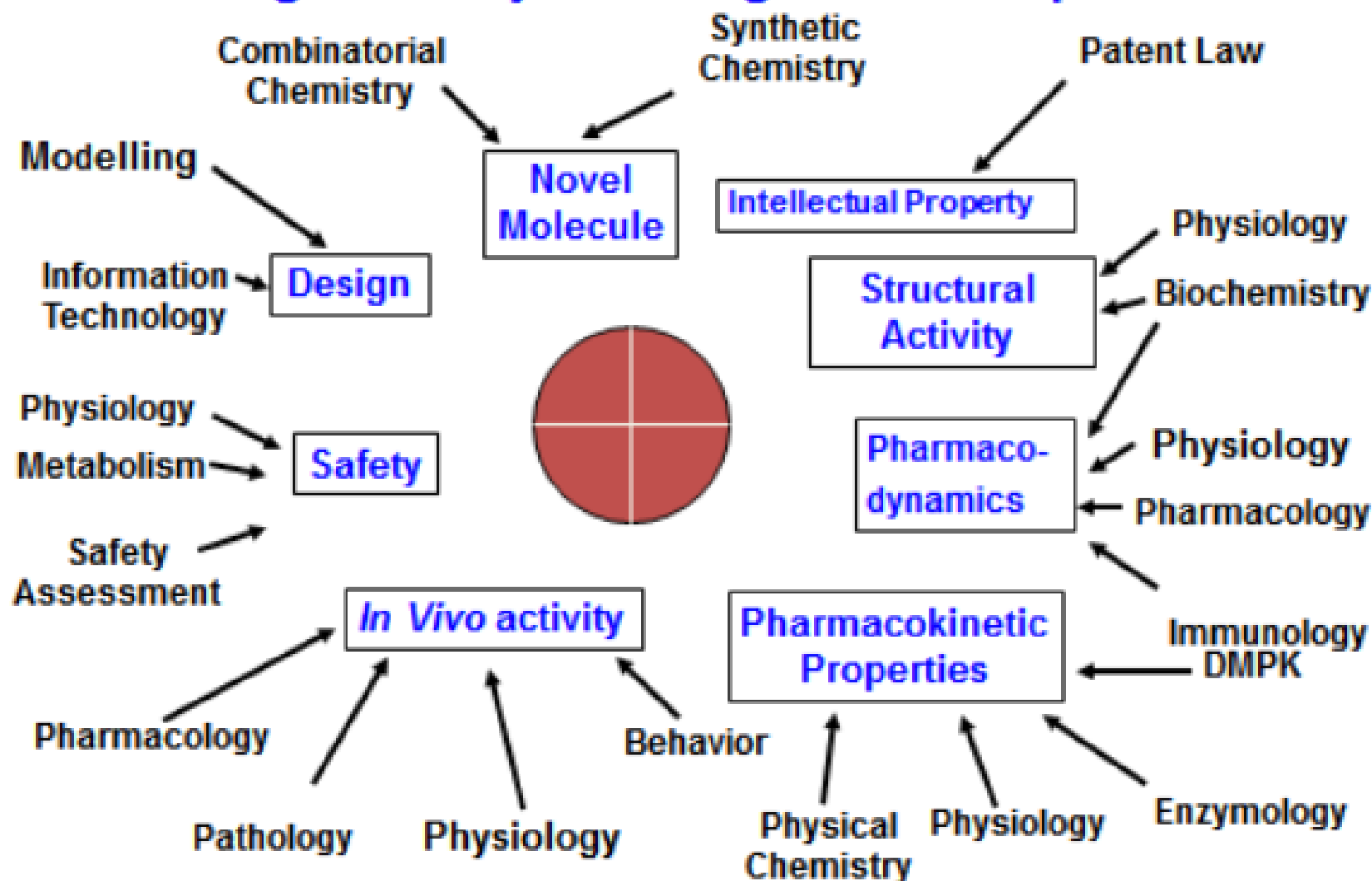
Medicinal Chemistry?

- Discovery, Design of new therapeutic chemicals
- Development of such into useful medicines
- The molecular basis of drug action.

Areas (Scope) of Study

- ❑ synthesis of new compounds
- ❑ isolation from natural sources
- ❑ structure-activity relationships (SAR & QSAR)
- ❑ absorption, transport, distribution & metabolic transformations and etc.
- ❑ elucidating the mode of **interaction** with **targets** & their **outcomes** (receptors/enzymes/DNA, etc...)

Drug Discovery—Convergence of Disciplines



MedChem_I_Pharma

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Definitions of Terms

- **New drug** is a drug that has not been declared safe and effective by qualified experts under the conditions prescribed, recommended, or suggested in the label and that may be a new chemical agent
- **Drug product (Dosage form; Finished product)** is one or more drug substances (active pharmaceutical ingredients) usually with excipients
- **Pharmaceutical formulation** is the processes in which different chemical substances i.e., active chemical substances will combined together to produce a medical compound i.e., medical drug
- **High-throughput screening** is a highly automated robotic system that tests small amounts of large numbers of compounds against potential targets.
- **Combinatorial Chemistry** is a process of generation of active compounds (hits) from a library of building blocks

- **Drug design** is the approach of finding a new drug by design based on the knowledge of its biological target. The process of drug discovery involves the identification of candidates, synthesis, characterization, screening and assays for therapeutic efficacy.
- Drug design involves the design of molecules that are complementary in shape and charge to the biomolecular target with which they interact and therefore will bind to it.
- **Drug discovery** is the process by which new drug candidates are discovered. The drug discovery process can be described as the identification and validation of a disease target and the discovery and development of a chemical compound to interact with that target. This interaction can be to block, promote or otherwise modify the activity of the target.
- **Drug development** is the process of bringing a new pharmaceutical drug to the market once a lead compound has been identified through the process of drug discovery.

- Drug development includes preclinical research on microorganisms and animals, filing for regulatory status for an investigational new drug to initiate clinical trials on humans, and may include the step of obtaining regulatory approval with a new drug application to market the drug.
- In a broader sense, the definition encompasses the entire process of drug discovery and clinical testing of novel drug candidates
- *In vitro*: In an artificial environment, as in a test tube or culture media
- *In vivo*: In the living body, referring to tests conducted in living animals
- *Ex vivo*: Usually refers to doing the test on a tissue taken from a living organism.

- **Drug delivery** is the method or process of administering a pharmaceutical compound to achieve a therapeutic effect in humans or animals. Its technologies are patent protected formulation technologies that modifies drug release profile, absorption, distribution and elimination for the benefit of improving product efficacy & safety and patient convenience & compliance
- **Structure-activity relationship (SAR)** is the relationship between chemical structure and pharmacological activity for a series of compounds.
- **Lead compound** is a compound that has a desirable biological activity with therapeutic relevance, but typically has some shortcoming that is likely to be overcome through the development of analogs.

What are drugs in the first place?

How does a medicinal chemist define drugs?

Drugs: are compounds that interact with a biological system to produce a biological response.



Definition, classification and nomenclature of drugs

Definition

- A **drug**, broadly speaking, is any substance that, when absorbed into the body of a living organism, alters normal bodily function.
- There is no single, precise definition, as there are different meanings in drug control law, government regulations, medicine, and colloquial usage.
- A drug is "a chemical" substance used in the treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being
- In medicinal chemistry, the chemist attempts to design and synthesize a medicine or a pharmaceutical agent which will benefit humanity according to the above mentioned definition.

Classification

- Drugs can be classified according to various criteria:

1. By origin—sources of drugs

A. Synthetic: Most of the drugs in use today are synthetic in origin. Such drugs are chemically pure and it is easy to maintain supply line.

Ex. : Aspirin, Paracetamol.

B. Natural: There are number of natural sources.

- Drugs may be obtained from 1. Plants 2. Animals 3. Minerals or 4. Microorganisms. The drugs may also be semi synthetic or synthetic compounds. The sources of drugs are summarized as follows:
- (a) Plants: A number of plant based drugs such as vincristine, taxol, digoxin, quinine, reserpine, ergotamine, ephedrine, colchicine etc. are still a part of standard therapy.
- (b) Animal: Some modern drugs continue to be derived from animal sources because the synthesis of such chemicals is very cumbersome and expensive. Ex. : Gonadotrophins, heparin, insulin, thyroid extracts and enzymes.

- (c) Microorganisms: Following the accidental discovery of penicillin from a mould in 1928 and its successful use in chemotherapy in 1940, a large number of antibiotics have been discovered from a variety of soil fungi and some bacteria. Ex.: Penicillin, Streptomycin, Tetracycline.
 - (d) Minerals: Minerals or mineral-containing medicated *springs have been in use since* time immemorial. Ex. iron, calcium, magnesium, aluminium, sodium, potassium etc.
- C. Semi-synthetic:** In some cases, especially with complex molecules, the synthesis of a drug may be very difficult or expensive and uneconomical. At the same time, the ones derived from natural sources may be impure. In these cases semi-synthetic processes are used.
- Ex. : 6-Aminopenicillanic acid
- Recently, bacteria have been enlisted as chemists to produce complex natural products, e.g., recombinant insulin by genetic engineering.
 - The technology is called **Recombinant DNA Technology**.

2. By Pharmacological action

- These drugs mainly affect the normal dynamic processes of the body. They are; Analgesics, Diuretics, Bronchodilators, Hypnotics, Antivirals, Antimalarials, Anti-arrhythmics, Antianginals, Vasodialators, Anti-hypertensives, Cardiotonics, Hypocholesteric agents, Antiallergic agents, Drugs acting on GIT etc...

3. By Chemical structure

- Drugs are classified according to the chemical moiety or functional group.
Ex. Steroids, Alkaloids, Quinolones, Glycosides, Betalactams, Halogenated compounds, *Alcohols, Carboxylic acids, Phenols, Nitro compounds, Amides, Sulphonamides, sulphones,etc.*

4. By their target (site) of action

e.g. Adrenergics, Dopaminergics, Cholinergics.....

Drug-Body Interactions

- The interactions between a drug and the body are conveniently divided into two classes
- **Pharmacodynamics**
 - “what the drug does to the body”
 - MOA
 - Effects
- **Pharmacokinetics**
 - “what the body does to the drug”
 - ADME (**A**bsorption, **D**istribution, **M**etabolism and **E**limination)

Factors affecting ability of drugs to reach active site

- There are certain vital factors that govern the ability of a drug to reach the active site soon after its administration through various modes known to us. These factors essentially include **absorption, distribution, biotransformation (metabolism) and elimination**. However, in all these instances, the drug molecule has to cross a few biological membrane in one form or the other.
- **Absorption** is the passage of the drug from its site of administration into the blood.
- **Distribution** involves the transport of drug molecules within the body. After the drug is absorbed into the bloodstream, it is carried by the blood or tissue fluids to its sites of pharmacologic action
- **Metabolism** is method by which drugs are inactivated or biotransformed by the body
- **Elimination** refers to the excretion of a drug from the body

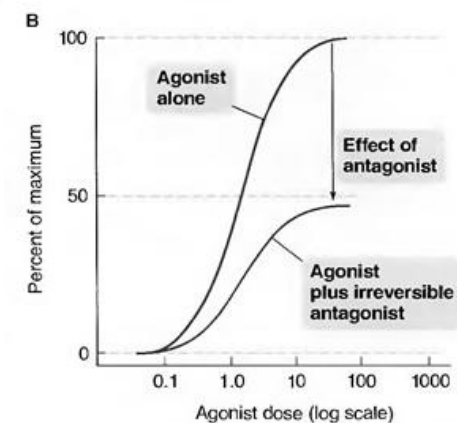
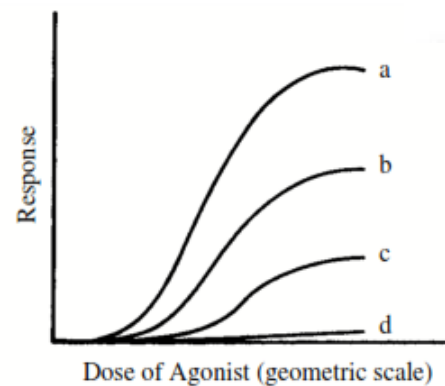
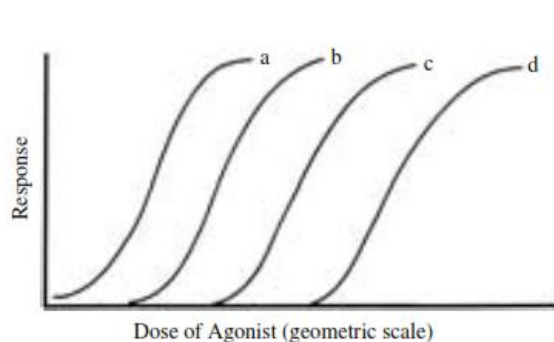
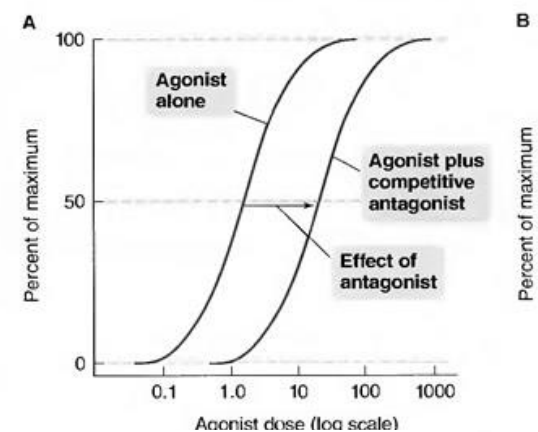
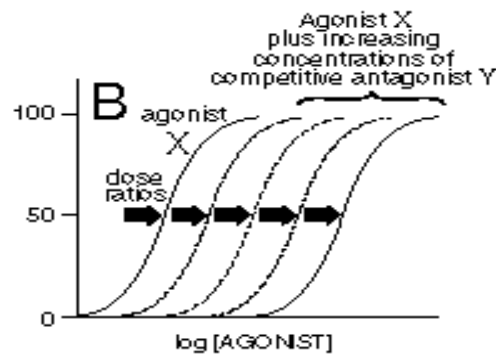
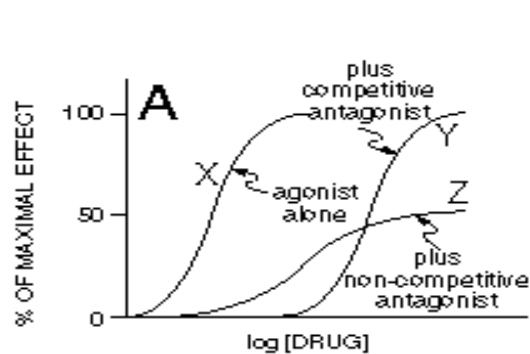
Drug-receptor interactions

1. Agonist: A drug that activates a receptor is known as an agonist, which has the following properties;

- Agonists can differ in both affinity and efficacy for the receptor
- High efficacy agonists are full agonists because they elicit maximal effects
- Low efficacy agonists are partial agonists because they cannot elicit a maximal effect at receptors even at high concentrations (false transmitters)
- Direct agonists act on receptors, while indirect agonists facilitate the actions of the endogenous agonist (the neurotransmitter, itself)

2. Antagonist: A drug that does not activate the receptor is an antagonist, which possesses the following features ;

- Antagonists also prevent the activation of the receptor by an agonist, thus antagonists are essentially zero efficacy drugs
- Competitive antagonists bind to the same binding site as the agonist and therefore compete with the agonist for that binding site
- Non-competitive antagonists have a different binding site to the agonist and therefore do not compete with the agonist. Some non-competitive antagonists have a binding site within the ion channel associated with the receptor complex.



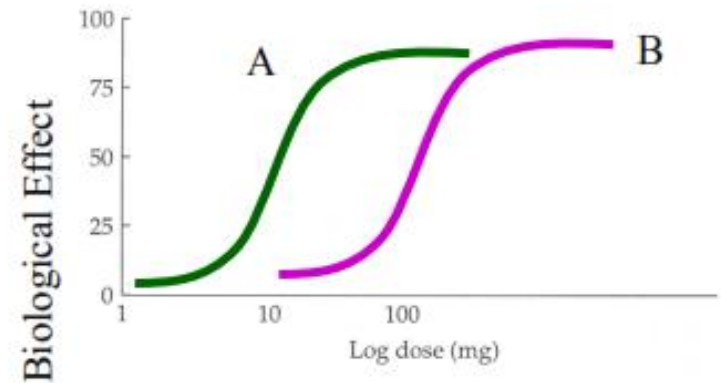
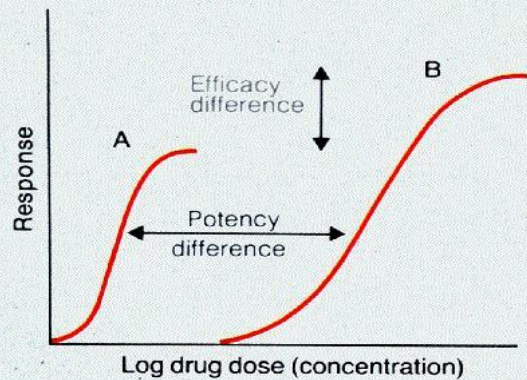
Idealized dose–response curves of an agonist in the absence (a) and the presence (b, c, d) of increasing doses of competitive antagonist.

Idealized dose–response curves of an agonist in the absence (a) and the presence (b, c, d) of increasing doses of a non competitive antagonist

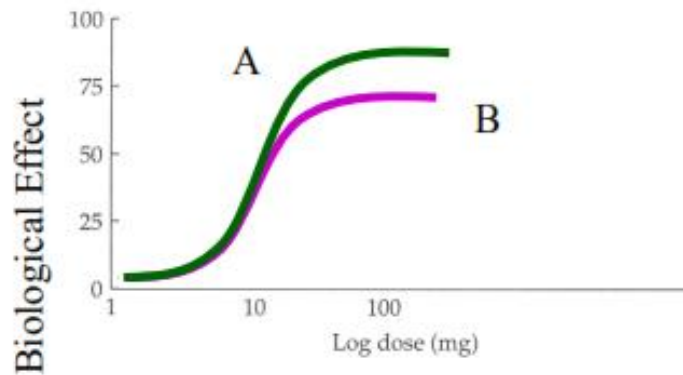
Response vs. concentration graphs

- Dose-response data are typically graphed with the dose or dose function (eg, $\log_{10}$ dose) on the x-axis and the measured effect (response) on the y-axis.
- **Efficacy** is a maximum effect that a drug can produce regardless of dose (greatest attainable response)
- **Potency** is an amount of a drug that is needed to produce a given effect
 - The lower the dose needed to produce a response, the more potent the drug
- **Bioavailability** is the proportion of the a drug or other substance that enters the circulation when introduced into the body and so it is able to have an active effect

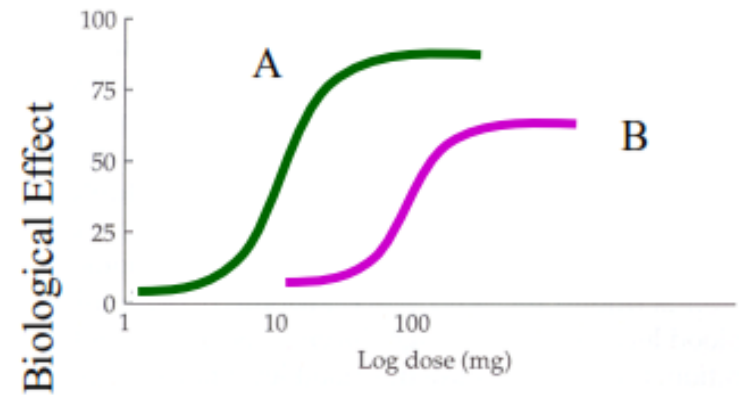
The difference between 'potency' (affinity) and 'efficacy' (activity)



Efficacy: $A=B$
Potency: $A>B$

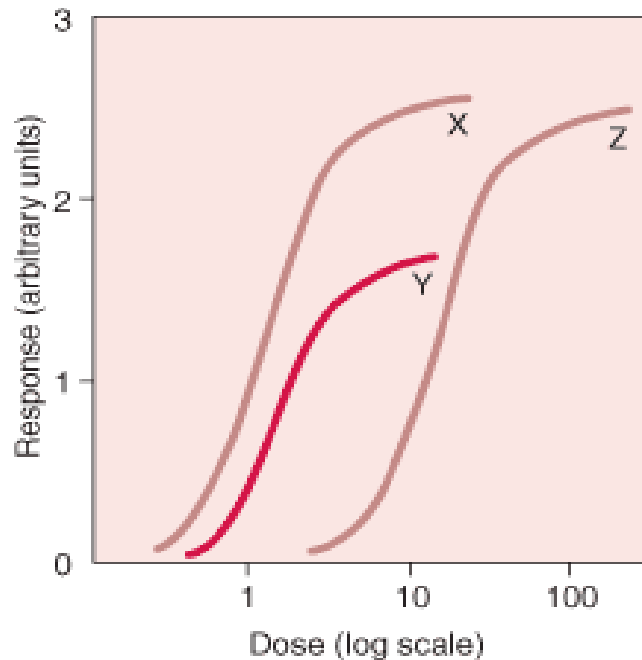


Efficacy: $A>B$
Potency: $A=B$

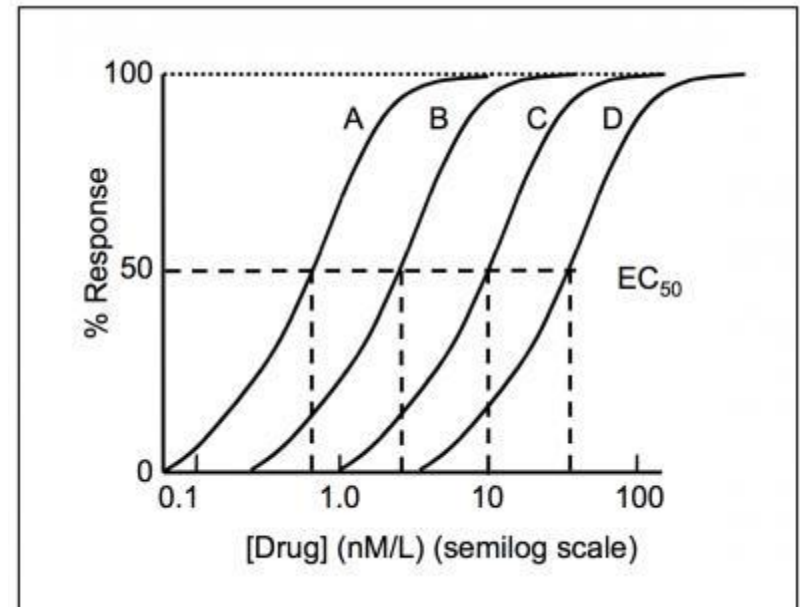


Efficacy: $A>B$
Potency: $A>B$

Compare the potency and efficacy of the following hypothetical dose-response curve.



Curve A



Curve B

- **Therapeutic index (TI)** describes a relationship between the doses of a drug that causes lethal or toxic effects with the dose that causes therapeutic effects. It is also called as therapeutic ratio.

$$\text{Therapeutic Index} = \text{LD50/ED50}$$

- LD50 is the minimum amount of drug that causes adverse effects in 50% of the population. LD50 could also be replaced with Toxic dose (TD50)
- ED50 is the quantity of a drug that can produce desired therapeutic effects in 50% of the population. Such types of studies are usually conducted in animal models.
- A high Therapeutic Index (TI) is preferable for a drug to have a favorable safety profile

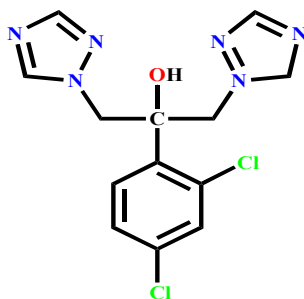
Nomenclature

■ Drugs have 3 or more names:

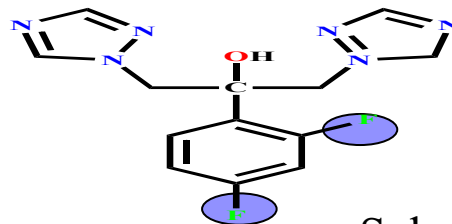
1. Code number or code designate
2. Chemical name
3. Proprietary, trial, **brand**, patented or trade name/mark
4. Non proprietary, **generic**, official or common name

i. Code Number

- Formed with the initial(s) name of a laboratory, the chemist or research team that prepared or tested the drug the first time and followed by a No. to represent drug.
- It does not identify the chemical nature or structure of the drug.
- It is discontinued as soon as an adequate name is chosen.



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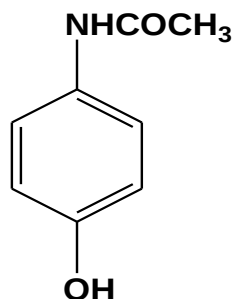


Fluconazole

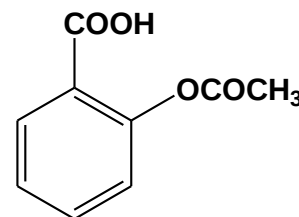
Substituent varied
Less toxic

ii. Chemical name

- Are the scientific (IUPAC) names, based on the molecular structure of the drug
- They describe the substances chemically. i.e., They depict the chemical/molecular structure of drugs in terms of atoms and molecules, and accompanied by a diagram of the chemical structures of drugs
- They are useful for chemists or technical personnel as they provide the precise arrangement of atoms and atomic groups in the molecule but are long and difficult and also not suitable for common use in prescribing drugs by physicians and also not used to identify the drug in a clinical or marketing situation .



N-(4-hydroxyphenyl)-acetamide
(Paracetamol)



2-Acetoxybenzoic acid
(Aspirin)

iii. The proprietary name, trade name, brand name ®

- These are names given to the drug that is discovered, developed by pharmaceutical company, and are their properties or trade marks of the manufacturing company
- These names are patented or copyrighted by terms selected by a manufacturer to designate a particular product. i.e., Once a new drug (usually known as pioneer drug) is discovered, the company files for a patent (20-25 years) to protect against other companies making copies and selling the drug.
- It should be written with capitalization of the first letter of each word of the name.

eg. Panadol® /Tylenol®, Vermox® /Quemox®

4. Generic, Common, Official, nonproprietary name

- Refers to the common established name, which a drug is known as an isolated substance, irrespective of its manufacturer.
- It should be simple, concise, and meaningful but often it is not.
- The first letter is not capitalized. Eg. paracetamol, aspirin
- It is chosen by official agencies, such as the U.S. Adopted names (USAN) council, sponsored by American Medical Association (AMA), the American Pharmaceutical Association (APA), the U.S. Pharmacopeial (USP) convention and the U.S. Food and Drug Administration (FDA).
- The World Health Organization (WHO) is the official agency to select, approve and disseminate names of drugs.

Genesis of Drugs or How are New Drugs Discovered?

- The procedure followed in drug design involves:
 1. The search for the **lead structures** or **prototype** (a well-established chemical substance of known biological activity)
 2. **Molecular manipulations** aimed for optimizing pharmacokinetic and pharmacodynamic (PD-PK) properties of the leads
 3. Design of suitable **pharmaceutical forms and dosage regimens**.

The discovery of new drugs or lead structure can be achieved mainly by one of the following routes:

A. Serendipity

- Accidental discovery of drugs in the laboratories or clinics by pharmacists, physicians or other investigators.
- E.g. Antibacterial action of penicillin was first observed by Fleming (1929) in a culture of bacteria that was contaminated by a fungus.
- The anti-inflammatory, analgesic and antipyretic activities of phenylbutazone were discovered while it was being used solely as a solubilizing agent for aminopyrine.

B. Random Screening

- All available chemical substances are submitted to a variety of biological tests in the hope that some may show useful activity.
- Not very rewarding.
- E.g. It is estimated that for a new anticonvulsant discovery through this process it might be necessary to screen 500,000 chemical compounds.

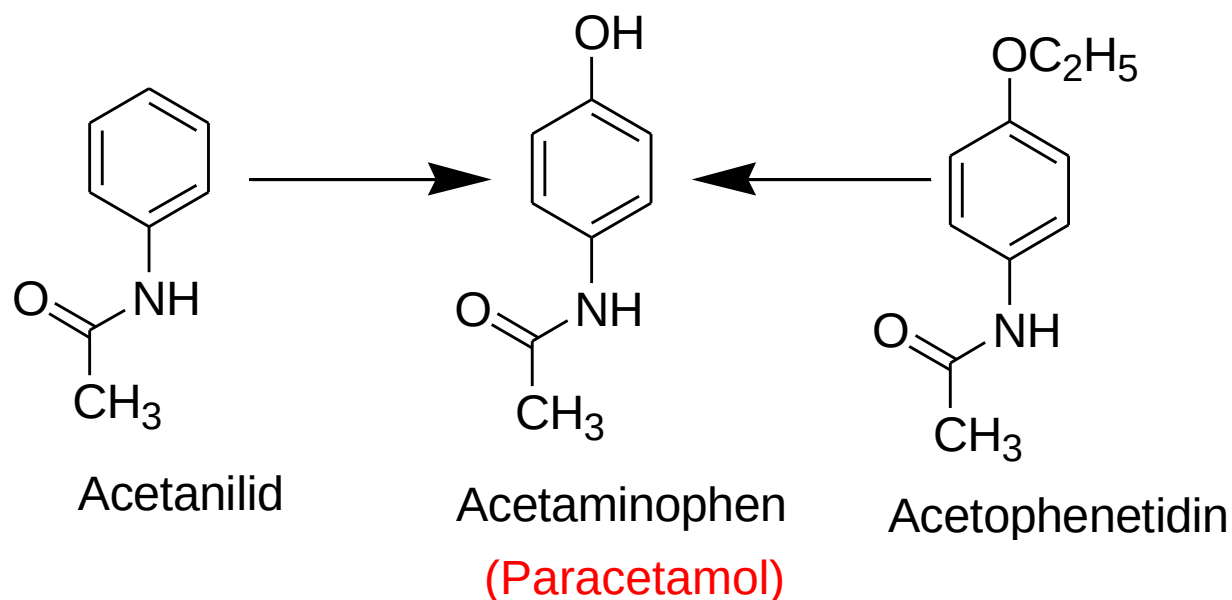
C. Rationally directed random screening e.g. the search for new antibiotics from microorganisms after the discovery of penicillin result in the recognition of more than 600 antibiotics, more than 100 of which are used in medicine. [See Fig 1](#)

D. Natural Products as Leads for Drug Discovery

- Morphine, atropine, ephedrine, curare and many others have binding affinities to certain receptors, enzymes, etc., and are thus useful as lead structures for drug discovery.

E. Drug Metabolites as Leads for Drug Discovery

- For example the acetanilide and acetophenetidin are metabolized to acetaminophen, which exerts the main analgesic actions.



F. Clinical Observation of Side Effects of Drugs

- Innumerable examples can be found in the literature, such as the hypoglycemic effect of some antibacterial sulfonamides, the antidepressant effect of ionized and the hypertensive effect of β -blocking agents.

G. Rational Approaches

- Rational approaches to drug design have now become the major routes to lead discovery.
- The key information that permits rational approaches to drug design is knowledge of the etiology of a given diseases or at least the biochemical processes that are disturbed.
- Examples of the rational approach are the discovery of the inhibitors of the angiotensin-converting enzyme and histamine H_2 -receptors antagonists

Molecular Modification or Manipulation

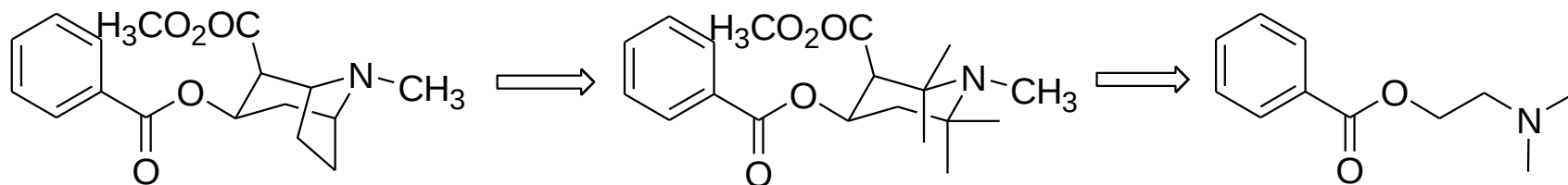
- The objectives of lead structures modification is to obtain drugs having more desirable properties than the prototype in potency, toxicity, specificity, duration of action, ease of application, stability and cost of production
- Once the lead compound is in hand two main aspects can be differentiated in molecular manipulation
 - ✓ General Process
 - ✓ Special Process

I. General Process

- Two general processes may be used in the method of variation

1. Disjunction or molecular simplification

- It is the systematic synthesis and evaluation of the simpler and simpler analogs of the lead compound.

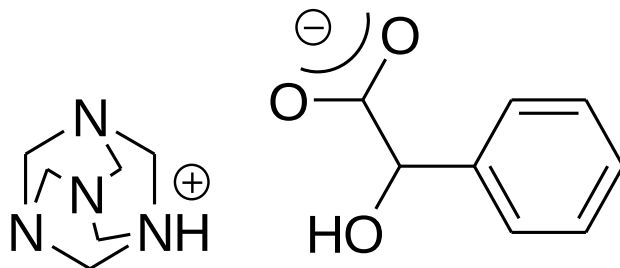


2. Conjunction or Molecular Association

- It is the synthesis and evaluation of more and more complex analogs of prototype. These analogs incorporate certain or all features of the lead compound.
- Three main types of association can be distinguished:

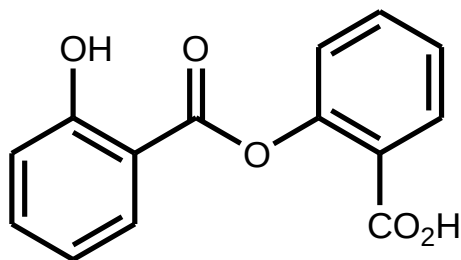
i. Molecular addition:

- Association of different moieties through weak forces such as electrostatic attraction and hydrogen bonding, example is methenamine mandelate.

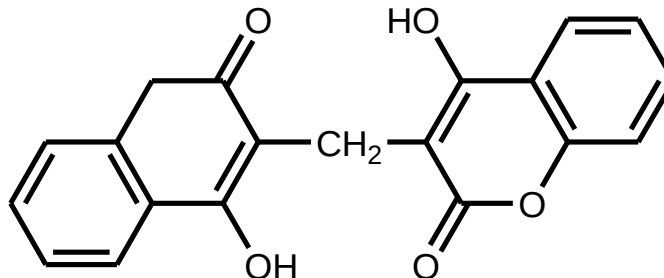


ii. Molecular Replication:

- Association of identical moieties through covalent bond formation (Identical twin drug) e.g., salicyl salicylate. This is also can be found among several natural products e.g., dicumarol.



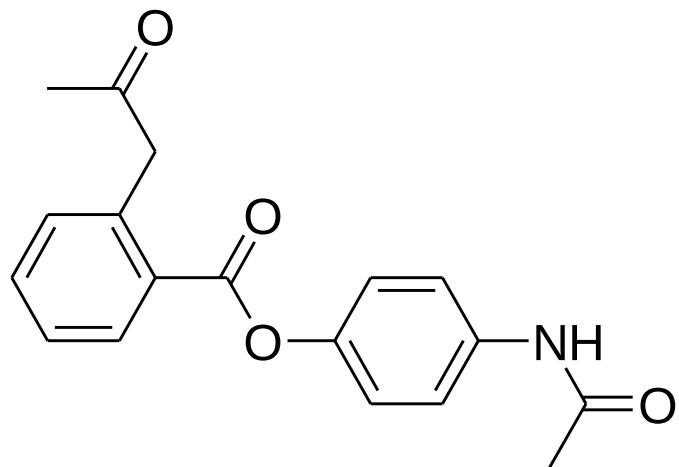
Salicyl salicylate



Dicumarol (anticoagulant)

iii. Molecular Hybridization:

- Association of different or mixed moieties through covalent bond formation (non-identical twin drug) e.g., acetaminosalol (aspirin with acetaminophen).



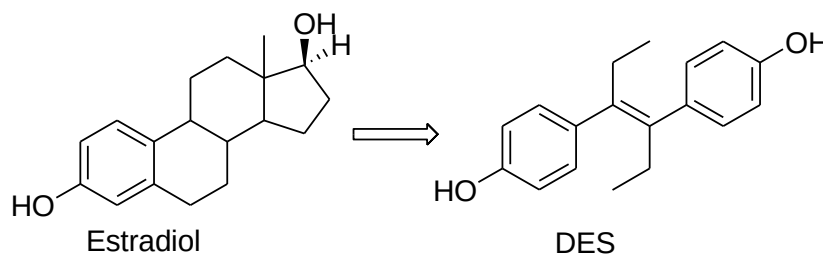
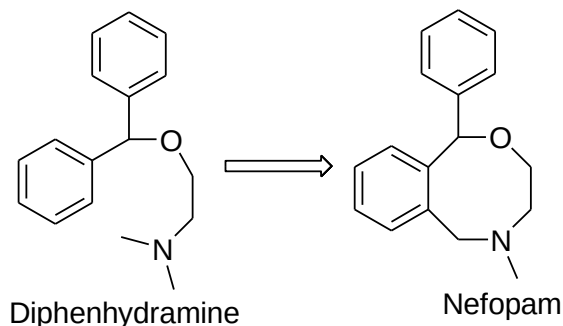
II. Special Process

- Several special processes for molecular modifications were grouped in two classes

A. Alterations which increase or decrease the dimensions and flexibility of a molecule, this can be achieved through the following:

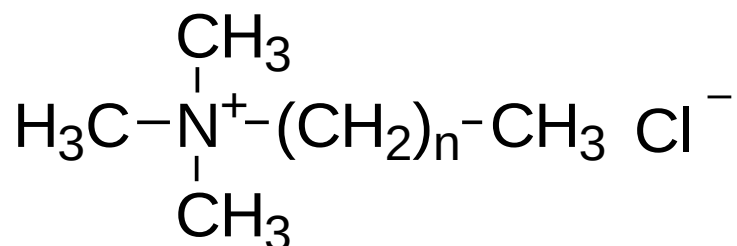
i. Ring closer or opening:

- Several examples are found of new drugs designed by either closure of a chain (e.g. nefopam from diphenhydramine) or opening of a ring (diethylstilbesterol from estradiol)



ii. Formation of Lower or Higher Homologues:

- A homologous series is analogs that differ in their structure by simple increment of in molecular formula.
- These may be produced by sequential chemical changes, which include increasing or decreasing the length of a carbon chain.
- For example, alkyltrimethylammonium analogs possess different types of activity depending on the length of alkyl groups:



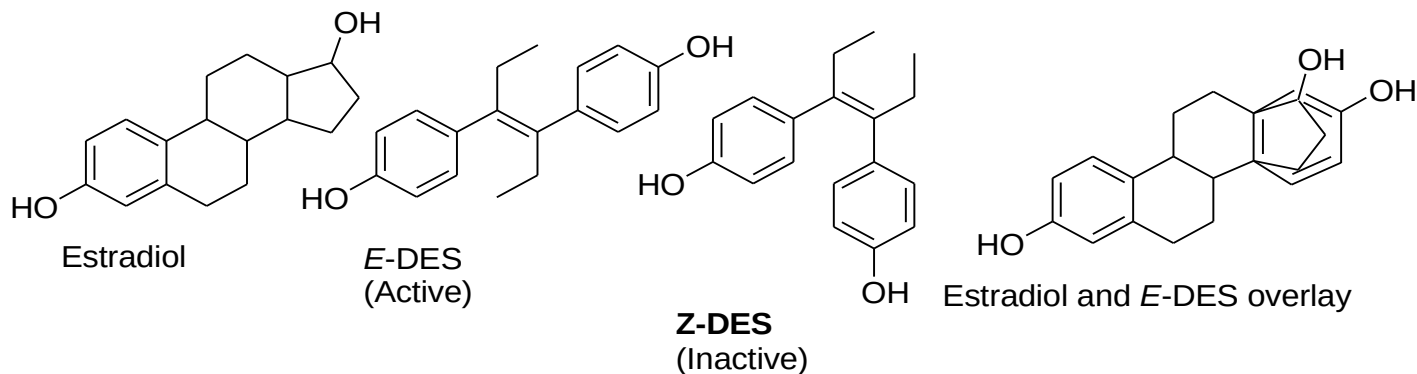
- If alkyl group is up to 6 carbons ($n = 5$) the compounds are muscarinic agonists. With 7 to 8 carbons ($n = 6$ or 7), these compounds are partial agonists. However, when the length is greater than 9 carbons ($n = 8$) these compounds are muscarinic antagonists.

iii. Introduction of Double Bonds:

- The introduction of double bonds may cause two main effects:
 1. The possibility of the existence of geometrical isomers with different activity

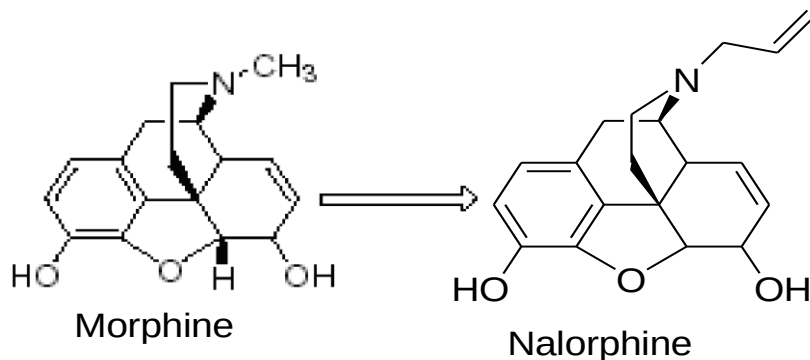
e.g., unsaturated propylamine H₁-antagonists, will be discussed later.
- The synthetic estrogen diethylstilbestrol (**DES**) is a tetra substituted ethane and exists in cis-trans forms.
- The trans isomer is much more active than the cis isomer. The reason is that the trans geometric isomer resembles the estrogen (estradiol) structure and better fits the estrogen receptor (designed by nature to fit estrogens) to bring about estrogen-like response.

- Here the shape of trans-DES positions the two hydroxyl groups for hydrogen bonding to the receptor.



2. Altering physicochemical properties

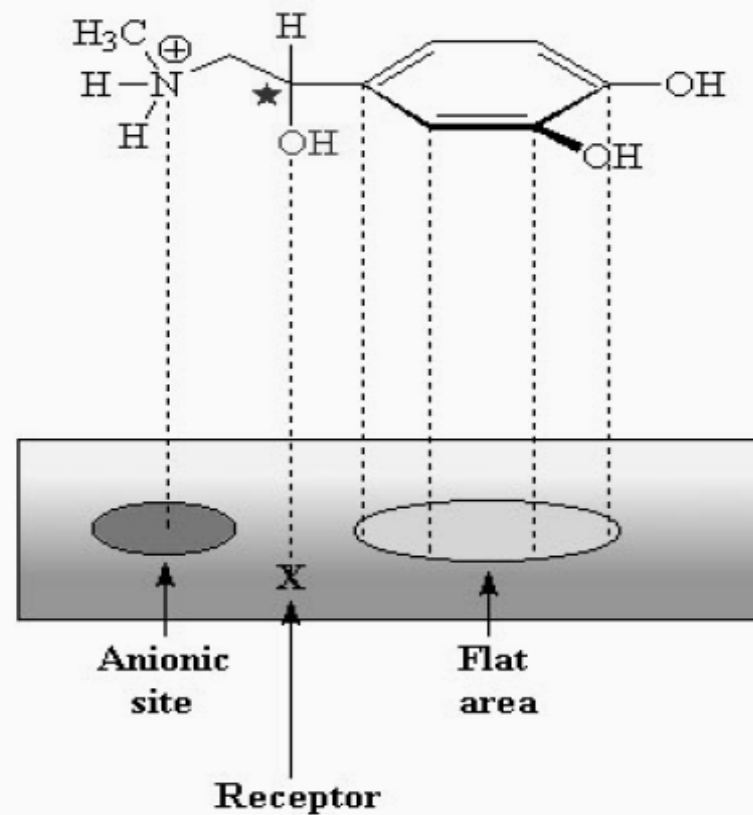
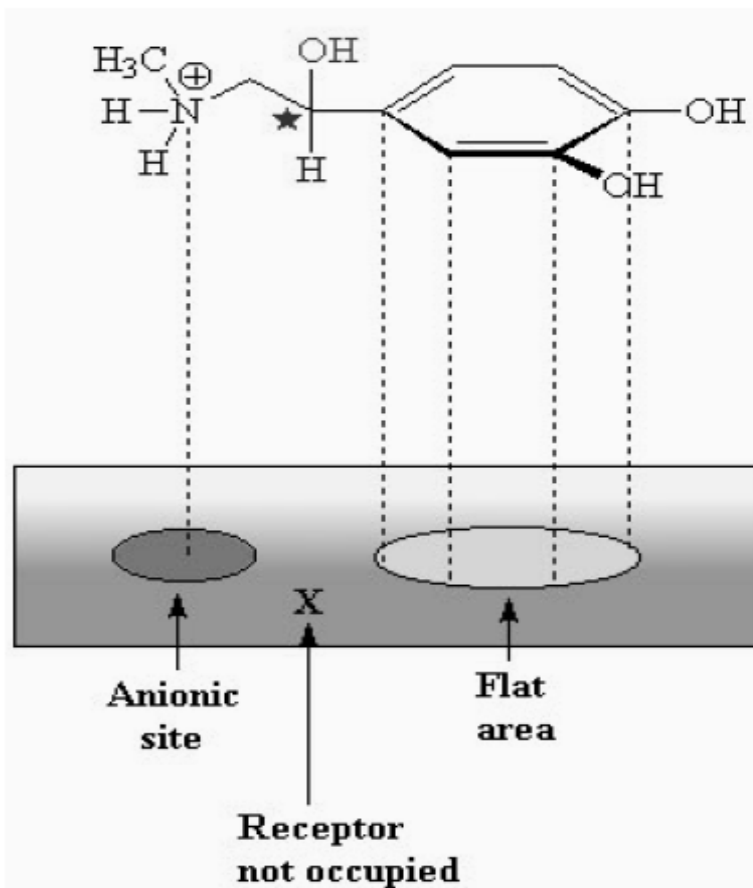
- Another example is the conversion of morphinic receptor agonists into the corresponding antagonists in case of morphine.



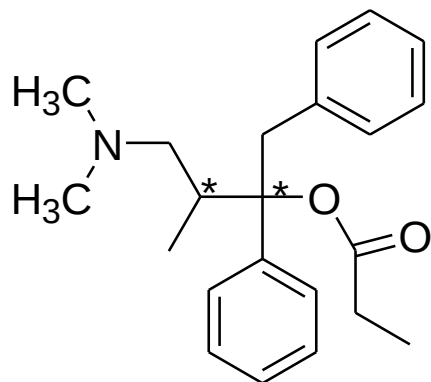
iv. Introduction of Chiral centers

- Receptors are chiral entities and the interactions of many drugs at specific sites exhibit chirality of interaction.
- They clearly prefer synthetic compounds with absolute configuration corresponding to the natural mediator. Thus changing the stereochemistry of the drug molecule may significantly alter its pharmacological activity.
- For example, the *R*(-)-isomer of epinephrine is more potent on both α - and β -adrenergic receptors than the *S*(+)-isomer.

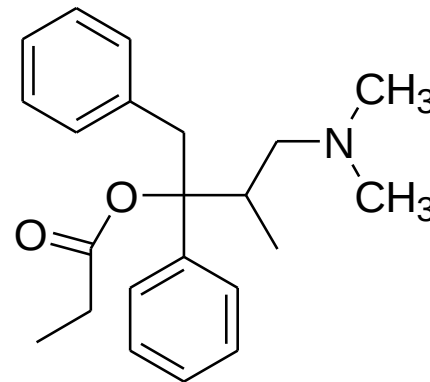
- The binding of epinephrine isomers and epinine (the desoxy analog of epinephrine) is illustrated below.
- Only *R* isomer can bind to all three sites, while both the *S* isomer and the deoxy isomer, which show similar activity, can bind to only two of these sites.



- Enantiomers may have different therapeutic activities as well.



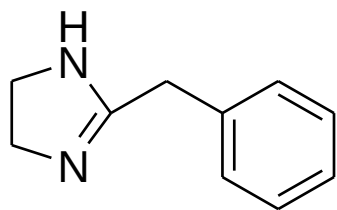
(+)-Dextropropoxyphene



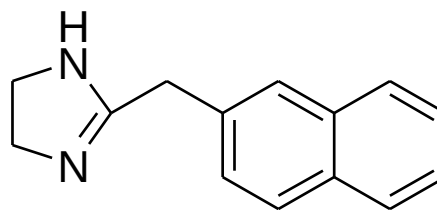
(-)-Levopropoxyphene

v- Introduction, Removal, or Replacement of Bulky groups:

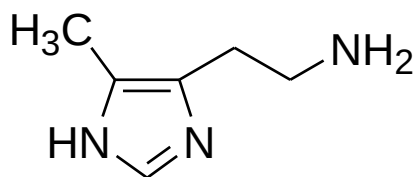
- This approach is used mainly to convert agonists into antagonists and vice versa as illustrated by the following examples:



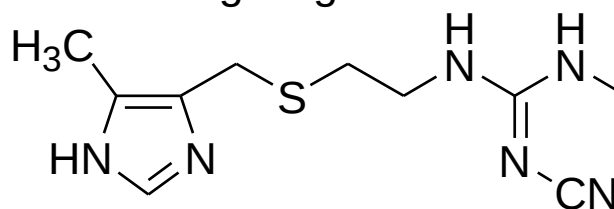
Tolazoline
 α -Adrenergic antagonist



Naphazoline
 α -Adrenergic agonist



4-Methylhistamine
 H_2 -receptor agonist



Cimetidine
 H_2 -receptor antagonist

- Another interesting example is found in β -lactamase resistant penicillin due to the introduction of bulky groups to position 6 of penicillanic acid.

B. Other Possible Modifications

i- Isosteric Replacements:

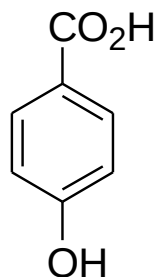
- *Isosteres* are molecules or groups of atoms that have the same number and arrangements of electrons.

For example, N_2 and CO , N_3^- and NCO^- . These substances have similar physical properties.

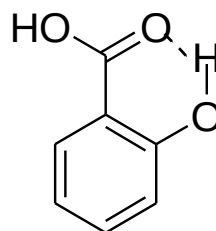
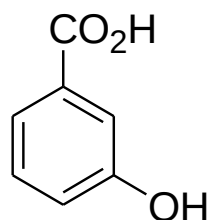
- *Bioisosterism* or nonclassical isosteres are groups or molecules which have chemical and physical similarities producing broadly similar biological properties.
 - In other words, they are compounds which fit the broadest definitions for isosteres and have the same type of biological activity.

ii- Change of Position or Orientation of certain groups:

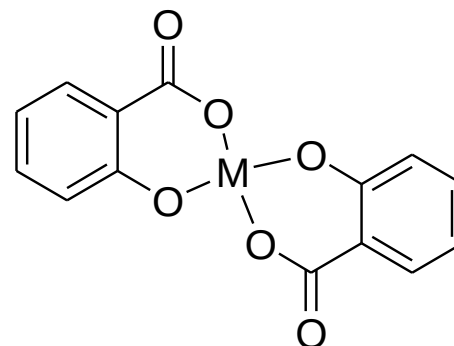
- The position of certain groups is sometimes essential for a given biological activity.
- Eg. Of the three isomers of hydroxybenzoic acid only the o-hydroxy is active, because it can form an intramolecular hydrogen bond and in this way, it can act as a chelating.



Inactive

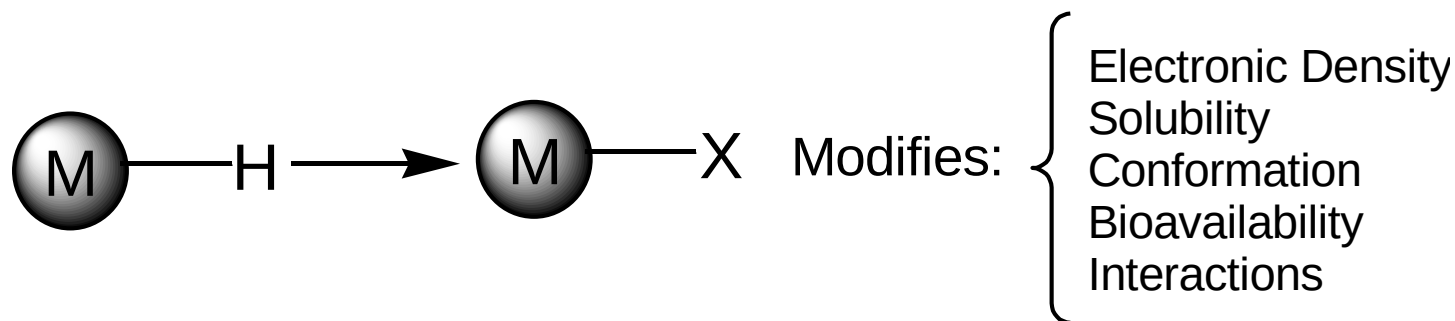


Active



iii- Modification toward Inhibition or Promotion of Various Electronic states:

- Certain chemical groups produce two important electronic effects,
- ✓ Inductive and conjugative.
- These effects can alter deeply physical and chemical properties and therefore, biological activity.



Structure-Activity Relationships (SAR)

- Basically, activity of a drug either increases or decreases as a result of some structure change (relative to the prototype).
- A Structure Activity Relationship (SAR) is a statement of the effect of structure change on biological activity within a congeneric series (a family) of compounds.
- SAR is the relationship between the chemical or three-dimensional structure of a molecule and its biological activity.
- The analysis of SAR enables the determination of the chemical groups responsible for evoking a target biological effect in the organism.
- This allows modification of the effect or the potency of a bioactive compound (typically a drug) by changing its chemical structure.
- Medicinal chemists use the techniques of chemical synthesis & computational drug design to insert new chemical groups into the biomedical compound and test the modifications for their biological effects.
- This method was refined to build mathematical relationships between the chemical structure and the biological activity, known as quantitative structure-activity relationships (QSAR)

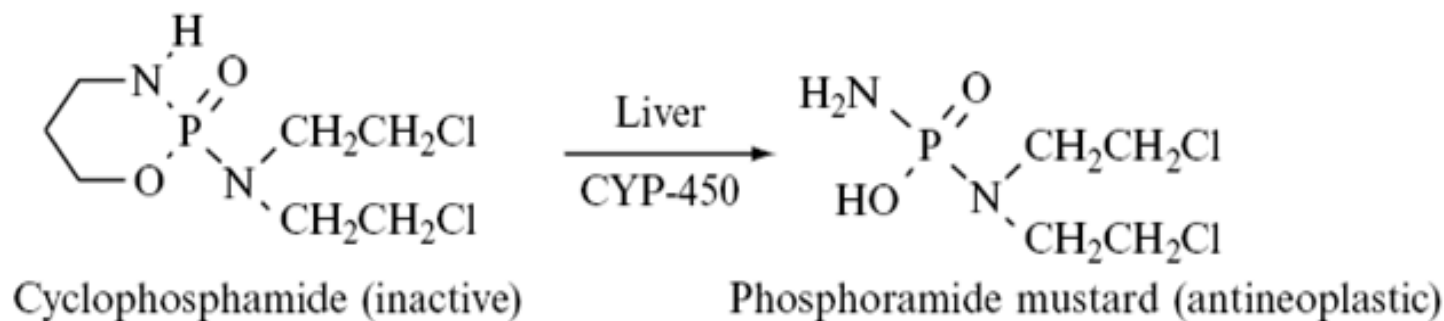
Introduction to prodrug design

Prodrugs

- Prodrugs are compounds that are biologically inactive but are metabolized to an active metabolite, which is responsible for the drug's action.
 - They are classified as either **bioprecursor** or **carrier prodrugs**.
 - Prodrugs may be designed to improve absorption, improve patient acceptance, reduce toxicity and also for the slow release of drugs in the body.

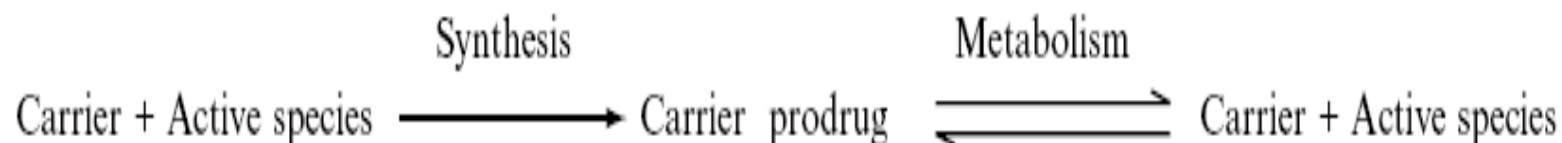
Bioprecursor prodrugs

- Bioprecursor prodrugs are compounds that already contain the embryo of the active species within their structure.
- This active species is liberated by metabolism of the prodrug

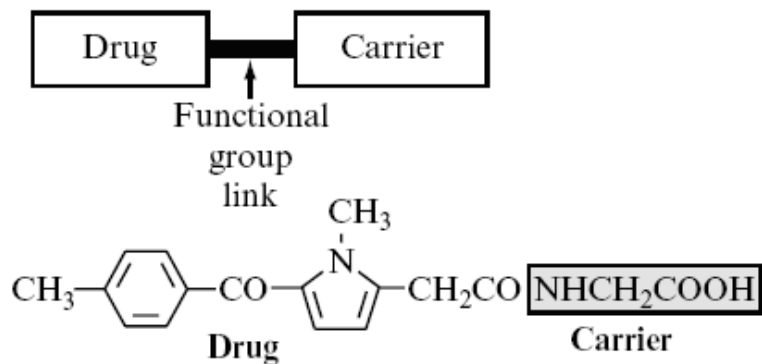


Carrier prodrugs

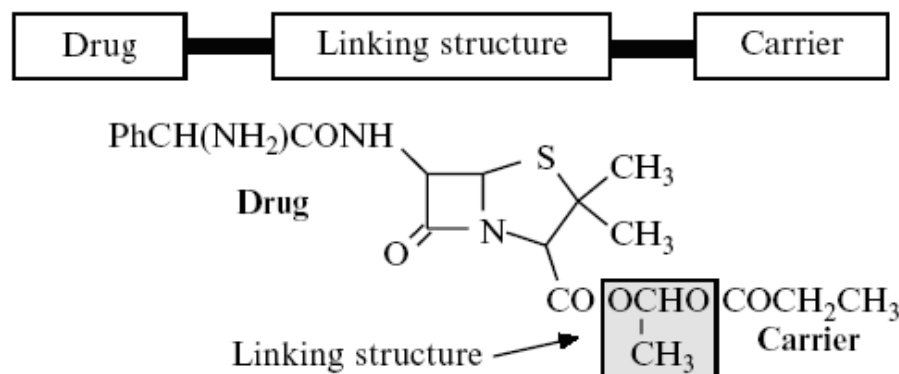
- Carrier prodrugs are formed by combining an active drug with a carrier species to form a compound with the desired chemical and biological characteristics, for example, a lipophilic moiety to improve transport through membranes.
- The link between carrier and active species must be a group, such as an ester or amide, that can be easily metabolized once absorption has occurred or the drug has been delivered to the required body compartment.
- The overall process may be summarized by:



- Carrier prodrugs that consist of the drug linked by a functional group to the carrier are known as **bipartate prodrugs** (Figure below).
- Tripartate prodrugs** are those in which the carrier is linked to the drug by a link consisting of a separate structure.



(a) Tolmetin-glycine prodrug



(b) Bacampicillin, a prodrug for ampicillin

Examples of (a) bipartate and (b) tripartate prodrug systems

The application of prodrugs

- Prodrugs are converted into the active drug within the body through enzymatic or non-enzymatic reactions. The various applications of prodrug approach are

A. Improving absorption and transport through membranes

- The transport of a drug through a membrane depends largely on its relative solubility in water and lipids.
- Good absorption requires that a drug's hydrophilic–lipophilic nature is in balance. The lipophilic nature of a drug may be improved by combining a lipophilic carrier with a polar group (s) on the drug.

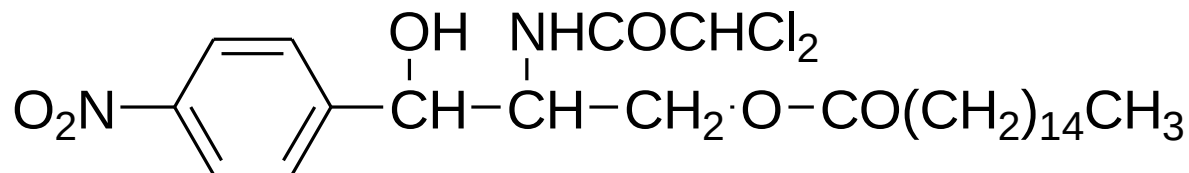
- However, it is difficult to select a lipophilic carrier that will provide the degree of lipophilic character required. If the carrier is too lipophilic, the prodrug will tend to remain in the membrane.
- Similarly, improving the water solubility of a drug may be carried out by introducing a carrier with a water solubilizing group or groups.

Table Examples of the reactions used to improve the lipophilic nature of drugs

Functional group	Derivative
Acids	an appropriate ester
Alcohols and phenols	an appropriate ester
Aldehydes	acetal
Ketones	acetal (ketal)
Amines	quaternary ammonium derivatives, amino acid peptides and imines

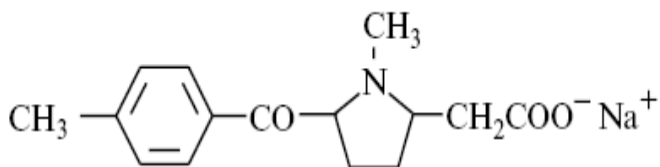
B. Improving patient acceptance

- Odour and taste are important aspects of drug administration.
- A drug with a poor odour or too bitter a taste will be rejected by patients, especially children.
- Furthermore, a drug that causes pain when administered by injection can have a detrimental effect on a patient.
- The formation of a carrier prodrug can sometimes alleviate some of these problems. For example, palmitic acid and other long chain fatty acids are often used as carriers, since they usually form prodrugs with a bland taste.

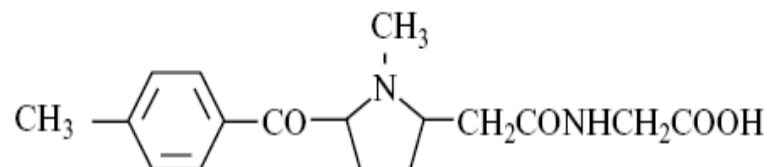


C. Slow release/increase duration

- Prodrugs may be used to prolong the duration of action by providing a slow release mechanism for the drug.
- Slow release and subsequent extension of action is often provided by the slow hydrolysis of amide and ester linked fatty acid carriers.
- Hydrolysis of these groups can release the drug over a period of time that can vary from several hours to weeks.
- For example, the use of glycine as a carrier for the anti-inflammatory tolmetin sodium results in the duration of its peak concentration being increased from about one to nine hours.



Tolmetin sodium

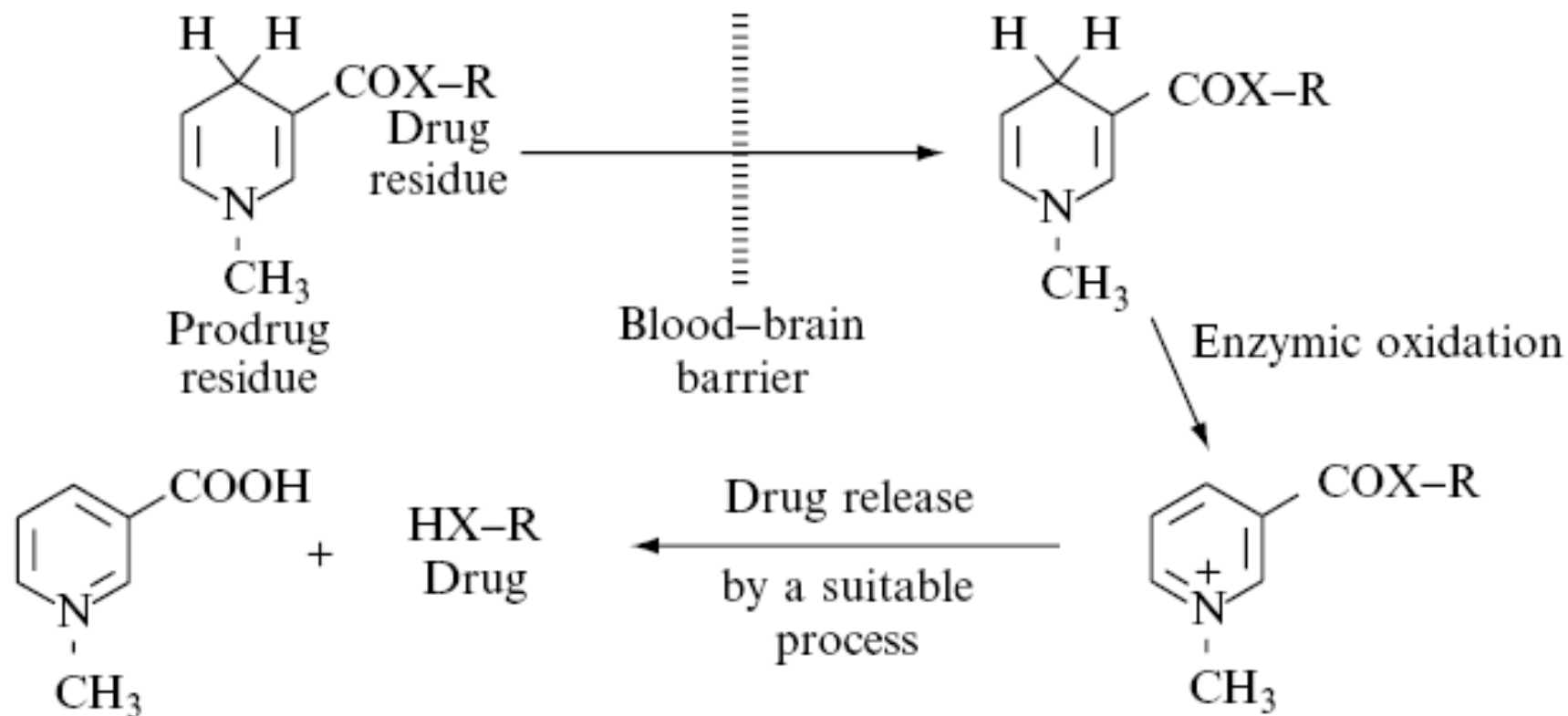


Tolmetin-glycine prodrug

D. Site specificity

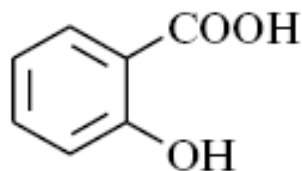
- In theory, it should be possible to design a carrier prodrug that would only release the drug in the vicinity of its site of action.
- Furthermore, once released, the drug should remain mainly in the target area and only slowly migrate to other areas. In addition the carrier should be metabolized to nontoxic metabolites.
- Unfortunately, these requirements have only been achieved in a few cases.
- One area where the site specific carrier prodrug approach has been used with some degree of success is to design drugs capable of crossing the blood–brain barrier.

- This barrier will only allow the passage of very lipophilic molecules unless there is an active transport mechanism available for the compound.
- A method developed by Bodor and other workers involved the combination of a hydrophilic drug with a suitable lipophilic carrier, which after crossing the blood–brain barrier would be rapidly metabolized to the drug and carrier.
- Once released, the hydrophilic drug is unable to recross the blood–brain barrier.
- The selected carrier must also be metabolized to yield nontoxic metabolites.

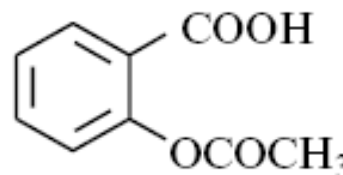


E. Minimizing side effects

- Prodrug formation may be used to minimize toxic side effects. For example, salicylic acid is one of the oldest analgesics known.
- However, its use can cause **gastric irritation** and **bleeding**.
- The conversion of salicylic acid to its **prodrug aspirin** by acetylation of the phenolic hydroxy group of salicylic acid improves **absorption** and also **reduces the degree of stomach irritation**, since aspirin is mainly converted to salicylic acid by esterases after absorption from the GI tract.
- This reduces the amount of salicylic acid in contact with the gut wall lining.



Salicylic acid



Aspirin

Computer Aided Drug Design (CADD)

- Computer aided drug design (CADD) a key process used in drug discovery to identify hits from compound libraries that may become leads for medicinal chemistry optimization. It comprises a broad range of theoretical and computational approaches that are part of modern drug discovery.
- Refers to the application of informatics methods with in rational drug design, to discover, design and optimize biologically active compounds.
- The parameters expected from a drug are safety, efficiency, stability, solubility, synthetic viability and novelty

-
- The two major types of drug design are ligand-based drug design and structure-based drug design

Ligand-based drug design (LBDD) (indirect drug design)

- relies on knowledge of molecules that bind to the biological target of interest or it is an approach used in the absence of the receptor 3D information
- The first category is about “finding” ligands for a given receptor.
- A large number of potential ligand molecules are screened
- It saves synthetic effort to obtain new lead compounds.

Structure-based drug design (SBDD) (direct drug design)

- relies on knowledge of the three dimensional structure of the biological target obtained through methods such as x-ray crystallography or NMR spectroscopy or SBDD is the one where you design your molecules according to the target, here u know the structure of the receptor
- First techniques to be used in drug design.
- Helped in the discovery process of new drugs.
- Information about the structural dynamics and electronic properties about ligands are obtained from calculations.

Why do we need new drugs?

- New drugs are constantly required:
 1. To combat drug resistance
 2. For improving the treatment of existing diseases
 3. For the treatment of newly identified diseases
 4. For the production of safer drugs by the reduction or removal of adverse side effects

Drug Resistance or Tolerance occurs when a drug is no longer effective in controlling a medical condition. Drug Resistance arises in people for a variety of reasons:

1. development of an enzyme that metabolises the drug
2. downregulation of receptors (repeated stimulation of a receptor results in the receptor being broken down)
3. appearance of a drug-resistant strains of microorganisms

Drug discovery and development: is an expensive process.

- Two major phases are involved in creating a new drug:
 - *The discovery phase: First to identify what should be manipulated (i.e. a target protein), and second to find some suitable drug candidates that can alter the target.*
 - *Clinical testing: Modern drug development requires extensive and expensive testing in order to obtain the necessary governmental approvals.*

Phases/stages in drug discovery and development processes

1. Target identification
2. Lead compound identification/selection
3. Lead optimization
4. Pre-clinical trials
5. Clinical trials



Phases in Drug Discovery & Development

- Expensive and time consuming
 - Starts with the isolation from natural sources or synthesis of novel chemical compounds.
 - Steps
 - ❑ Target identification
 - ❑ Target validation
 - ❑ Lead compound identification/selection
 - ❑ Lead optimization
 - ❑ Preclinical test
 - ❑ Clinical test
 - ❑ Nomenclature
 - ❑ Registration
 - ❑ Manufacturing
-

Identification of biological target

- Drug target is a specific macromolecule, or biological system, which the drug will interact with (target could be proteins, genes ion-channel, nuclear receptor and RNA)

Lead compound identification

- A lead compound is the one that has basic structural requirements for exhibiting the desired action against the chosen target, but not yet good enough to be the drug itself. It is a compound that has a desirable biological activity with therapeutic relevance, but typically has some shortcoming that is likely to be overcome through the development of analogs to give a compound with further enhanced action/better action.

Lead optimization

- Lead optimization is the one of the most important steps in drug development. The chemical structure of the lead compound is used as a starting point for chemical modifications in order to improve potency, selectivity, or pharmacokinetic parameters.
 - At this stage, medicinal chemists conduct extensive SARs to improve potency and selectivity.
 - Also, this is the opportunity to improve physicochemical and drug-like properties

Preclinical test

- yields information on the biological effects of new substances.
 - to explore the drug's efficacy and safety before it is administered to patients
 - Animals and in vitro systems
 - Pharmacodynamics
 - Pharmacokinetics
 - toxicology and special toxicological studies
 - mutagenicity
 - carcinogenicity
 - Teratogenicity
-

- Acute toxicity- large single dose
- Sub chronic and chronic toxicity-repeated dose
- IND-approval by FDA to proceed to clinical trial

Clinical trials

- Testing of new drug on human
- 4 phases

Phase I

- Healthy volunteers (20-80)
- To determine whether effects observed in animal experiments also occur in humans.
- to establish the dose level at which signs of toxicity first appear.
- This phase evaluates the safety, tolerability and PK and PD of the new molecule.

Phase II

- ❑ the new drug is administered to patients for the first time(80-100).
 - ❑ Efforts are concentrated on evaluating efficacy and on establishing an optimal dose range
 - ❑ dose–response studies are a critical part of phase II studies.
 - ❑ Monitoring subjects for adverse effects is also an integral part of phase II trials.
-

Phase III

- ❑ On patients with increased number (hundreds to thousands)
 - ❑ The new drug will be compared with standard treatments in terms of therapeutic outcome.
 - ❑ to verify the efficacy of the drug and to detect effects that may not have surfaced in the phase I and II trials, during which exposure to the drug was limited.
 - ❑ NDA-to drug regulatory body
 - ❑ Approval of NDA
 - ❑ Manufacturing
-

Phase IV

- ❑ Post marketing surveillance
 - ❑ Observing long term effects of the drug on the general population after approval
 - ❑ Report from professional
 - ❑ Renewal or cancellation of licenses
-

QSAR in drug design

- QSAR is an attempt to **remove the element of luck** from drug design by establishing a mathematical relationship in the form of an equation between biological activity and measurable physicochemical parameters.
- These parameters are used to represent properties such as **lipophilicity**, **shape** and **electron distribution**, which are believed to have a major influence on the drug's activity. These equations may be used by the medicinal chemist to make a more informed choice as to which analogues to prepare.
- The main properties of a drug that appear to influence its **activity** are its, **lipophilicity**, the **electronic effects** within the molecule and the **size and shape** of the molecule (steric effects).

- **Lipophilicity** is a measure of a drug's solubility in lipid membranes. This is usually an important factor in determining how easily a drug passes through lipid membranes.
- **The electronic effects** of the groups within the molecule will affect its electron distribution, which in turn has a direct bearing on how easily and permanently the molecule binds to its target molecule.
- **Drug size and shape** will determine whether the drug molecule is able to get close enough to its target site in order to bind to that site.
- The parameters commonly used to represent these properties are **partition coefficients** for lipophilicity, **Hammett's constants** for electronic effects and **Taft's steric constants** for steric effects.

- QSAR derived equations take the general form:

$$\text{biological activity} = \text{function}\{\text{parameter(s)}\}$$

- In which the activity is normally expressed as $\log[1/(\text{concentration term})]$, usually C, the minimum concentration required to cause a defined biological response.

1. Lipophilicity

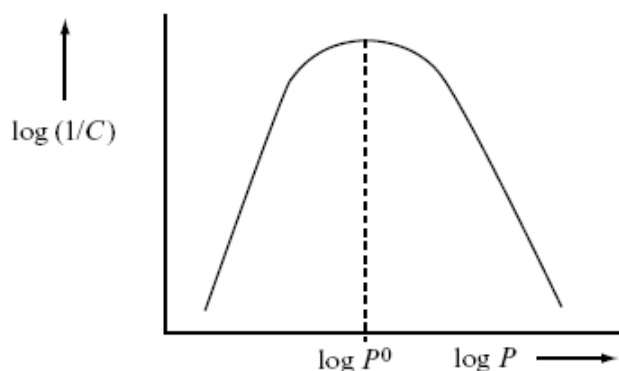
- Two parameters are commonly used to represent lipophilicity, namely the **partition coefficient** (P) and the **lipophilicity substituent constant** (p).
- The former parameter refers to the whole molecule whilst the latter is related to substituent groups.

1.1 Partition coefficient (P)

- Organic medium/aqueous system partition coefficients are the parameters to use as a measure of the ease of movement of the drug through the membranes.
- The n-octanol–water system is frequently chosen because it appears to be a good mimic of lipid polarity and has an extensive database.
- The relationship between P and drug activity **for a wide range of p values** is given by

$$\log(1/C) = -k_1(\log P)^2 + k_2 \log P + k_3$$

- Where k_1 , k_2 and k_3 are constants that are normally determined by regression analysis.



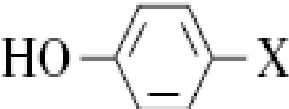
1.2 Lipophilic substituent constants (p)

- Lipophilic substituent constants are also known as hydrophobic substituent constants. They represent the contribution that a group makes to the partition coefficient and were defined by Hansch and co-workers by the equation:

$$\pi_X = \log P_X - \log P_H \quad P = \frac{[\text{drug}]_{\text{lipid}}}{[\text{drug}]_{\text{water}}}$$

- Where P_H and P_X are the partition coefficients of the standard compound and its substituted derivative respectively.
- However, when several substituent's are present, the value of p for the compound is the sum of the p values of each of the separate substituent's.
- The value of p for a specific substituent will vary with the structural environment of the substituent

Examples of the variations of π values with chemical structure

Substituent X	Aliphatic systems R-X			
-H	0.00	0.00	0.00	0.00
-CH ₃	0.50	0.56	0.52	0.49
-F	-0.17	0.14		0.31
-Cl	0.39	0.71	0.54	0.93
-OH	-1.16	-0.67	0.11	-0.87
-NH ₂		-1.23	-0.46	-1.63
-NO ₂		-0.28	-0.39	0.50
-OCH ₃	0.47	-0.02	0.18	-0.12

- A positive p value indicates that a substituent has **a higher lipophilicity than hydrogen** and so will probably increase the concentration of the compound in the n-octanol layer and by inference its concentration in the lipid material of biological systems.
- Conversely, a negative p value shows that the substituent has a **lower lipophilicity than hydrogen** and so probably increases the concentration of the compound in the aqueous media of biological systems.
- Lipophilic constants are frequently used when dealing with a series of analogues in which only the substituents are different.

- As an example, consider the log P values for benzene (log P = 2.13), chlorobenzene (logP = 2.84) and benzamide (logP = 0.64).

1. Calculate the hydrophobicity constant of Chlorine (π_{Cl}) in Chloro benzene?

2. Calculate the logP value for Chlorobenzamide if π_{CONH_2} is -1.49 and π_{Cl} is 0.71?

2. Electronic effects

- The distribution of the electrons in a drug molecule has a considerable influence on the distribution and activity of a drug.
- The first attempt to quantify the electronic effects of groups on the physicochemical properties of compounds was made by Hammett.

The Hammett constant (σ)

- The distribution of electrons within a molecule depends on the nature of the electron withdrawing and donating groups found in that structure.
- Hammett used this concept to calculate what are now known as Hammett constants (σ_x) for a variety of monosubstituted benzoic acids

$$\sigma_x = \log \frac{K_x}{K_H} = \log K_x - \log K_H.$$

$$\sigma_x = pK_H - pK_x$$

- Where K_H and K_x are the equilibrium constants for benzoic acid and monosubstituted benzoic acids respectively.

- Its value varies depending on whether the substituent is an overall electron donor or acceptor.
- A negative value for σ_X indicates that the substituent is acting as an electron donor group since $K_H > K_X$.
- Conversely, a positive value for σ_X shows that the substituent is acting as an electron withdrawing group as $K_H < K_X$.

Substituent	Hammett constants	
	σ_m	σ_p
–H	0.00	0.00
–CH ₃	–0.07	–0.17
–C ₂ H ₅	–0.07	–0.15
–Ph	0.06	–0.01
–OH	0.12	–0.37
–Cl	0.37	0.23
–NO ₂	0.71	0.78

3. Steric effects

- The first parameter used to show the relationship between the shape and size (bulk) of a drug, the dimensions of its target site and the drug's activity was the **Taft steric parameter (E_s)**.
- It was followed by Charton's steric parameter (n), Verloop's steric parameters and the molar refractivity (MR) amongst others.

The Taft steric parameter (E_s)

- Taft (1956) used the relative rate constants of the acid catalysed hydrolysis of α -substituted methyl ethanoates to define his steric parameter because it had been shown that the rates of these hydrolyses were almost entirely dependent on steric factors.
- He used methyl ethanoate as his standard and defined E_s as:

$$E_s = \log \frac{k_{(\text{XCH}_2\text{COOCH}_3)}}{k_{(\text{CH}_3\text{COOCH}_3)}} = \log k_{(\text{XCH}_2\text{COOCH}_3)} - \log k_{(\text{CH}_3\text{COOCH}_3)}$$

or $E_s = P^k_H - P^k_X$

Where k is the rate constant of the appropriate hydrolysis and the value of $E_s=0$ when $X = \text{H}$.

Examples of the Taft steric parameter E_s

Group	E_s	Group	E_s	Group	E_s
H–	1.24	F–	0.78	CH ₃ O–	0.69
CH ₃ –	0.00	Cl–	0.27	CH ₃ S–	0.19
C ₂ H ₅ –	–0.07	F ₃ C–	–1.16	PhCH ₂ –	–0.38
(CH ₃) ₂ CH–	–0.47	Cl ₃ C–	–2.06	PhOCH–	–0.33

Hansch analysis

- Hansch analysis attempts to mathematically relate drug activity to measurable chemical properties. It is based on Hansch's proposal that drug action could be divided into two stages:
 1. the transport of the drug to its site of action;
 2. the binding of the drug to the target site.
- Hansch postulated that the biological activity of a drug could be related to these parameters by simple mathematical relationships based on the general format:

$$\log 1/C = k_1(\text{partition parameter}) + k_2(\text{electronic parameter}) + k_3(\text{steric parameter}) + k_4$$

- Where C is the minimum concentration required to cause a specific biological response and k1, k2, k3 and k4 are numerical constants obtained by feeding the values of the parameters selected by the investigating team into a suitable computer statistical package.

What types of molecules in the cell are affected by drugs?

- The main molecular targets affected by drugs are:
- ✓ **Lipids:-**Cell membrane lipids
- ✓ **Proteins:-**Receptors, Enzymes, Carrier proteins, Structural proteins (tubulin)
- ✓ **Nucleic acids:-** DNA and RNA
- ✓ **Carbohydrates:-**Cell surface carbohydrates and Antigens and recognition molecules

There are four important attracting and binding forces...

- Covalent bonds
- Ionic bonds
- Hydrogen bonds
- Van der Waals bonds

Physico-chemical properties in relation to biological action

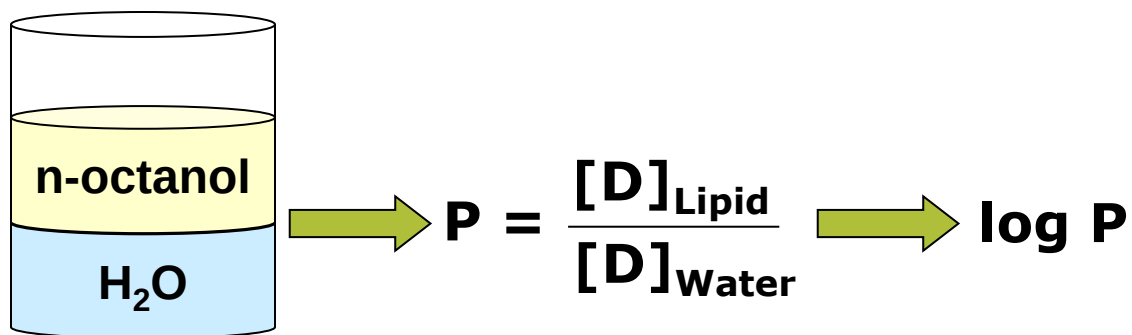
- At the most fundamental level, the ability of a chemical compound to elicit a pharmacological/therapeutic effect is related to the influence of various physical and chemical (physicochemical) properties of the chemical substance on the biomolecule(s) that it interacts with.
- Among the most pharmacologically influential physico-chemical properties of organic medicinal agents are;
- Solubility, Partition coefficient, Dissociation constant (pK_a), Hydrogen bonding, Molar refractivity (MR), Ionization, Drug shape, Complexation, Surface activity, Protein binding, Bioisosterism.

Lipinski's rule of five

- Lipinski's rule of five (RO5) is a rule of thumb to evaluate drug likeness or determine if a chemical compound with a certain pharmacological or biological activity has properties that would make it a likely orally active drug in humans.
- The rule is important to keep in mind during drug discovery when a pharmacologically active lead structure is optimized step-wise to increase the activity and selectivity of the compound as well as to insure drug-like physicochemical properties are maintained as described by Lipinski's rule.
- Candidate drugs that conform to the RO5 tend to have lower attrition rates during clinical trials and hence have an increased chance of reaching the market.

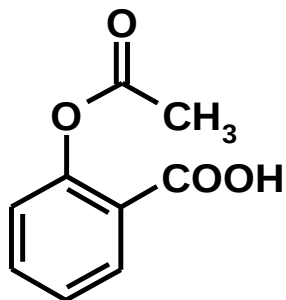
❑ **Lipinski's rule states that, in general, an orally active drug has no more than one violation of the following criteria:**

1. MW < 500
2. Fewer than five H-bond donating functions
(nitrogen or oxygen atoms with one or more hydrogen atoms)
3. Fewer than 10 H-bond accepting functions
(nitrogen or oxygen atoms)
4. Calculated logP (ClogP) between -1 and +5



Principles of Drug Design

❑ *Lipinski's Rule of Fives Examples*



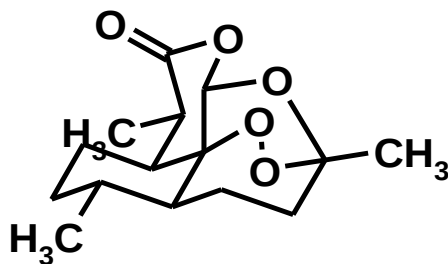
Aspirin

MW = 180 Da

H-bond donors = 1

H-bond acceptors = 4

ClogP = 2.44



Artemisinin

MW = 282 Da

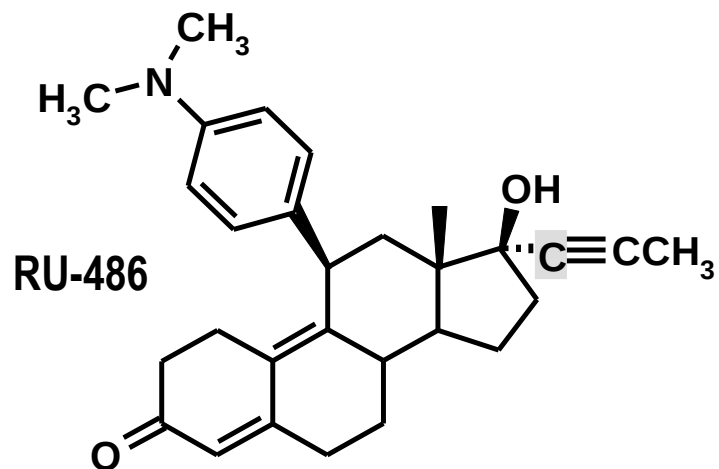
H-bond donors = 0

H-bond acceptors = 5

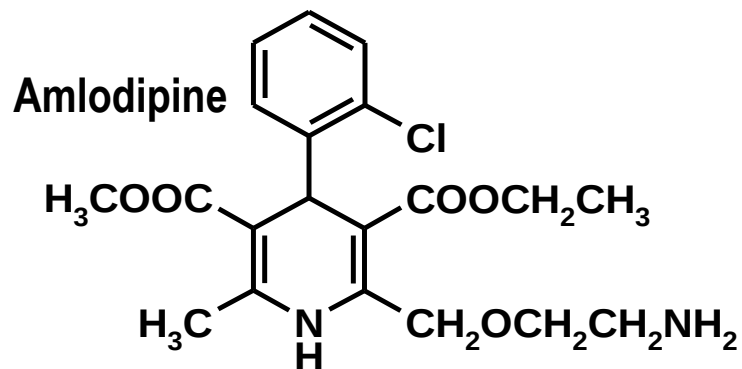
ClogP = 4.15

Principles of Drug Design

❑ *Lipinski's Rule of Fives Examples*



MW = 429 Da
H-bond donors = 1
H-bond acceptors = 3
ClogP = 4.74



MW = 407.5 Da
H-bond donors = 2
H-bond acceptors = 7
ClogP = -0.17

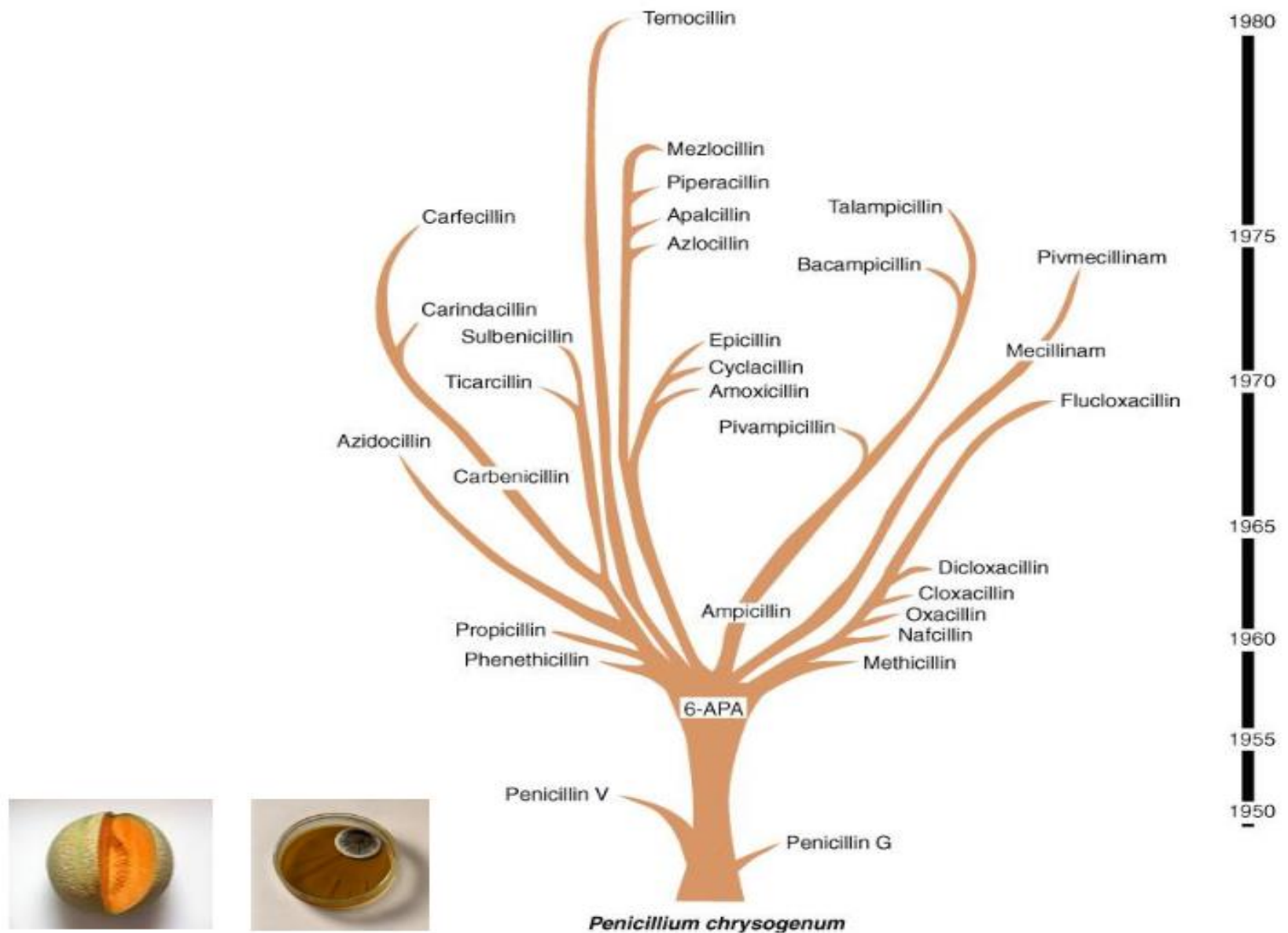


Figure 1